

Data Architecture Reference

A measured reference for data architecture and state design

Edition 1 (draft)

Built from commit repo/book-evidence-registry-v2-6-g9b04419 (9b044195316b62973520dd8e5d620e8f1141b2e4)
— working tree false.n Typst 0.15.1 (unknown commit) (image
sha256:032e292249bcd378480cc7c142cfa324b63ef8aadeb88d7e7230320c4c9c422f).n Evidence registry book/evidence/v2/
claims.json, digest 976e3635f5d2f769c3ab71e35e13083dc13bc728a82be32e10a7bfbe5f12e756.

1 Edition and provenance

This is a generated artefact: Typst source plus the evidence registry, compiled in a pinned Podman image. **Every claim in this book must resolve to a claim in the v2 registry**, and every number must resolve to a measured cell in a cited report. Nothing here is hand-entered at build time; the fields below are injected from the build script.

Source commit	9b044195316b62973520dd8e5d620e8f1141b2e4
Source describe	repo/book-evidence-registry-v2-6-g9b04419
Working tree at build	false
Built at (UTC)	2026-09-22T19:12:08Z
Typst version	0.15.1 (unknown commit)
Typst image digest	sha256:032e292249bcd378480cc7c142cfa324b63ef8aadeb88d7e7230320c4c9c422f
Evidence registry	book/evidence/v2/claims.json
Evidence digest	976e3635f5d2f769c3ab71e35e13083dc13bc728a82be32e10a7bfbe5f12e756
Source tree hash	c35e036e023604b5ee64e081a33afbf23c1dff91d8d58cf8dc5c0982a17348a2

The book is assembled with Typst inside a pinned Podman image; the evidence index at the end is rendered from the registry file at build time.

Coverage gap

Environment and provenance limits travel with every number: all results come from one host (host-zenbook-ux5406sa) with shared cores and no real network between “cluster” nodes. Run tags mark the **producing code state**, not a tree containing the results. A reader reproducing a run must check out the run tag to get the code, then read the committed results directory.

2 How to read a number here

- Every family, variant, topology and scenario section follows the same progressive structure: quick choice, where it thrives and perishes, common scenarios, mechanism, evidence and reproduction.
- A **direct** callout is measured in the cited run. An **analogy** callout is a labelled transfer to a family that was **not** measured. A **gap** callout is missing evidence.
- Gaps are never null results: “not measured” must not be read as “would not change this”.
- No cross-study numeric ranking appears anywhere; no comparability record exists.

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3 How to choose

CHAPTER · HOW TO CHOOSE

This chapter holds the decision path a reader follows before any family detail. Its prose is written by the HIGH book-synthesis-v1 task; the structure and the evidence hooks below are fixed here.

3.1 Answerability before performance

Direct evidence

The first question is what the schema must be able to **answer**, not how fast it answers. Two measured answerability failures make this a rule rather than a preference.

v2-07-refund-answerability-and-storage

append-only history / snapshots · single_run_directional

A ticketing database chosen only to prevent overbooking cannot report refunds at all: in 14 of 15 designs the refunds window is

4 Major families

CHAPTER · MAJOR FAMILIES

The six major families are fixed by the goal. Each section follows the same progressive structure; prose is written by book-synthesis-v1, evidence is attached here.

4.1 normalized facts

- 4.1.1 Quick choice in plain language
- 4.1.2 Where it thrives and where it perishes
- 4.1.3 Common scenarios
- 4.1.4 Reusable mechanism
- 4.1.5 Evidence and reproduction

v2-01-normalized-index-first

normalized facts · single_run_directional

In the charity -> person -> donation tree, the first large read win is the query index: D2 indexed scores 13.3x D1 minimal's read score; flattening the foreign key instead (D3 flat+FK) reaches 20.5x.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-01-study01-d2d3-copied-key-package](#)

Limits: PostgreSQL 1 node and YugabyteDB 1/3 nodes in one run; small scale; one trial per cell; read score is a blended measure, not one query

Source: [studies/01-charity-tree/reports/20260912-small.md](#) · run 20260912-small · tag none (legacy)

v2-06-hot-drop-designs

normalized facts · single_run_directional

For a hot drop, pre-created seats taken by compare-and-set or SKIP LOCKED (P3 CAS, P2 skip-locked) sell 4-8x faster than any single counter row, but only with retry backoff in mind, and a counter on the event row also blocks unrelated edits to it. Both deliberately wrong controls fired in the same run.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial per cell; the run has 42 cells of which 2 failed (C2 count (SER) on the two YugabyteDB topologies); those cells are not evidence; hot-drop workload only

Source: [studies/02-ticket-booking/reports/20260913T021206Z.md](#) · run 20260913T021206Z · tag run/02-ticket-booking/20260913T021206Z

v2-12-lock-vs-version-upper-bound

normalized facts · single_run_directional

Under one hot key with 16 writers a lock-before-update (c2_pessimistic_lock) delivered 1.76x the throughput of a version check (c1_optimistic_version), an UPPER BOUND on the lock's advantage because the run could not separate it from extra per-retry harness work; an unchecked read-modify-write (x1_lost_update_control) lost 93.7% of acknowledged updates.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one key, one installed product, 16 writers; PostgreSQL single node, small scale; one trial; -retries 1 control not run; 1.76x is one point on a writer curve, not a curve

Source: [studies/04-configuration-portal/reports/20260921T1215Z-small.md](#) · run 20260921T1215Z-small · tag run/04-configuration-portal/20260921T1215Z-small

4.2 rolldown / duplicated data

4.2.1 Quick choice in plain language

4.2.2 Where it thrives and where it perishes

4.2.3 Common scenarios

4.2.4 Reusable mechanism

4.2.5 Evidence and reproduction

v2-05-d3-read-advantage-repeated

rolldown / duplicated data · repeated_controlled

D3's (d3_flattened_fk) read advantage over the normalized indexed design survives five fresh loads; the recency SQL and actual heap avoidance explain the gain, while its insert cost remains imprecise.

Trials: 1 whole-run replications, 5 fresh-load trials per cell, 1 inner iterations.

Limits: five fresh-load trials per cell, one session; the insert-cost estimate is imprecise and must not be quoted as a ratio

Source: studies/01-charity-tree/reports/20260913T125342Z-v3.md · run 20260913T125342Z-v3 · tag run/01-charity-tree/20260913T125342Z-v3

v2-gap-05-colocation-unverified

rolldown / duplicated data · gap

Physical data placement is a common open gap: no study verifies actual node placement. Study 01 D7/D24 and Study 03 L3 are placement intent with measured effects, not verified colocation; Study 02 records its own placement gap; Study 04 y1/y2 and Study 05's placement pair were never executed.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: containers on one laptop are not evidence of colocation; physical placement is undocumented to today's standard, not proven uncolocated

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

4.3 rollups / materialized aggregates

4.3.1 Quick choice in plain language

4.3.2 Where it thrives and where it perishes

4.3.3 Common scenarios

4.3.4 Reusable mechanism

4.3.5 Evidence and reproduction

v2-02-rollup-family

rollups / materialized aggregates · single_run_directional

Parent rollups win the aggregate read score (D4 rollup/trg 73.8x and D5 rollup/app 69.9x against D1 minimal) and pay for it on inserts (2.6k/s and 2.1k/s against 6.2k/s).

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial per cell; PostgreSQL 1 node numbers shown; YugabyteDB in the same report; the write cost is per-insert on a fresh load

Source: studies/01-charity-tree/reports/20260912-small.md · run 20260912-small · tag none (legacy)

v2-04-recency-maintained-index

rollups / materialized aggregates · single_run_directional

The recency flag works, but an index on the rollup column answers the same question faster: comparing d20_recency_flag with d22_recency_rollup_idx in the same run gives the reported read speedup on PostgreSQL and YugabyteDB.

Trials: 1 whole-run replications, 3 fresh-load trials per cell, 1 inner iterations.

Confounds: conf-02-study01-d20d22-write-axis

Limits: the write comparison is confounded: d22 also maintains D4's multi-column rollup package, so its write cost is not the index alone; three trials per cell; claim-02's v1 trials field was recorded as 1 and is corrected here

Source: studies/01-charity-tree/reports/20260916T090036Z-v3.md · run 20260916T090036Z-v3 · tag run/01-charity-tree/20260916T090036Z-v3

v2-13-trigger-rollup-cost

rollups / materialized aggregates · single_run_directional

Maintaining the portal overview with a per-row trigger (n3_rollup_trigger) costs 6.85x on whole-configuration replacement (83.7 vs 573 ops/s, p99 2.5 -> 42 ms) while the application rollup (n4_rollup_app) keeps writes near the reference (593 ops/s) for the same ~15x read gain.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: PostgreSQL single node, small scale, one trial; the big effects (6.85x write, ~15x read) are above the run's ~1.7x read floor; smaller differences are not attributable; the trigger's product is a structural guarantee, not speed

Source: studies/04-configuration-portal/reports/20260921T1215Z-small.md · run 20260921T1215Z-small · tag run/04-configuration-portal/20260921T1215Z-small

4.4 embedded documents

4.4.1 Quick choice in plain language

4.4.2 Where it thrives and where it perishes

4.4.3 Common scenarios

4.4.4 Reusable mechanism

4.4.5 Evidence and reproduction

v2-03-embedding-not-ahead

embedded documents · single_run_directional

Embedding the donation child in the donor row (D6 embedded) never came out ahead of the normalized designs on reads: 2.4x D1 minimal's read score, below both indexed designs, with a 0.57x insert rate.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial; embedding here means a JSON/array child column, not a native document database

Source: studies/01-charity-tree/reports/20260912-small.md · run 20260912-small · tag none (legacy)

v2-11-section-seat-map

embedded documents · single_run_directional

On YugabyteDB the section document (L2 document) answers the 100k-seat map 4.5-11x faster than per-seat rows, while the section-sharded variant (L3 sharded) collapses point lookups to 26 ops/s; PostgreSQL shows no attributable difference. This is a family/package comparison, not a one-decision pair.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: conf-03-study03-layout-family-package

Limits: one read-only run, one trial, 5 s per measurement; quota-bound nodes; read-only phases; no control could fire; differences below ~1.8x are not attributable

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

4.5 append-only history / snapshots

4.5.1 Quick choice in plain language

4.5.2 Where it thrives and where it perishes

4.5.3 Common scenarios

4.5.4 Reusable mechanism

4.5.5 Evidence and reproduction

v2-07-refund-answerability-and-storage

append-only history / snapshots · single_run_directional

A ticketing database chosen only to prevent overbooking cannot report refunds at all: in 14 of 15 designs the refunds window is unanswerable and the sales/revenue and last-purchase reports are wrong across a refund. The append-only ledger (X1 CAS+ledger) that fixes it costs about 33-35% more storage.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: answerability is a schema fact, not a timing; the +33-35% storage delta is deterministic and stable across three runs; the P3 -> X1 read rows of this run are not attributable (cell-level effect) and are excluded

Source: studies/02-ticket-booking/reports/20260920T234953Z.md · run 20260920T234953Z · tag run/02-ticket-booking/20260920T234953Z

v2-08-ledger-race-cost-128-buyers

append-only history / snapshots · single_run_directional

At 128 buyers the ledger costs 1.1-2.3x of the sell-out race on PostgreSQL and 1.8-2.2x on YugabyteDB with tight spreads, 2.1-2.5x on the refund itself, and not one under-booked seat anywhere; it delays the organiser's own edit by up to 5x.

Trials: 1 whole-run replications, 3 fresh-load trials per cell, 128 inner iterations.

Confounds: [conf-05-study02-buyer-arms-disagree](#)

Limits: one small run, closed loop, quota-bound nodes; scoped to 128 buyers; must not be extrapolated to 32 buyers; single run, three trials per tier

Source: studies/02-ticket-booking/reports/20260921T205212Z.md · run 20260921T205212Z · tag run/02-ticket-booking/20260921T205212Z

v2-09-postgres-ledger-ratio-unsettled

append-only history / snapshots · single_run_directional

The two arms of the P3/X1 pair disagree in sign on PostgreSQL (32 buyers with P3 first: X1 faster; 128 buyers with X1 first: P3 faster), and the 32-buyer arm's own P3 trials disagree by 92-283%. No PostgreSQL ledger-cost ratio is quotable from this study; the run's own YugabyteDB range is 1.7-5.7x (not the headline's narrower 1.7-2.1x).

Trials: 1 whole-run replications, 3 fresh-load trials per cell, 32 inner iterations.

Confounds: [conf-05-study02-buyer-arms-disagree](#)

Limits: buyer count and design order changed together, so neither can be isolated; one arm is a partial cell on yb-cluster3 (two-trial medians); a repeated-design interleaved control is required before any PostgreSQL ratio is quoted

Source: studies/02-ticket-booking/reports/20260921T183722Z.md · run 20260921T183722Z · tag run/02-ticket-booking/20260921T183722Z

4.6 external read copies / caches

4.6.1 Quick choice in plain language

4.6.2 Where it thrives and where it perishes

4.6.3 Common scenarios

4.6.4 Reusable mechanism

4.6.5 Evidence and reproduction

v2-14-cache-throughput-gain

external read copies / caches · single_run_directional

Adding a cache to warm cacheable reads buys 2.2-3.5x in throughput and an order of magnitude in p99 (15.8-27.1 ms to 0.8-2.0 ms). This is an add-cache result under resource_framing db-only, not equal-total resource efficiency, and Redis here is a shared cache, never an authoritative key-value store.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-04-study05-db-only-cache](#)

Limits: one trial per measurement, one machine, closed loop; small scale (800 donors), PostgreSQL + Redis cache only; warm cacheable reads only; the Redis container's budget is outside the comparison (gap); no YugabyteDB, cluster, verified placement or open-loop demand

Source: studies/05-cache-consistency/reports/20260921T-survey3.md · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

v2-15-three-instance-staleness

external read copies / caches · mechanism_supported

Three logical application instances sharing one Redis cache on the write-through + relaxed strategy record about 87% wrong reads, while every single-instance relaxed cell records zero; a private in-memory lease cannot coordinate instances, and a database version token does not remove it. The 87.81% figure belongs to the OWNED model (o:opt:red:wth:rel); the legacy model's cell is 87.24%.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-04-study05-db-only-cache](#)

Limits: single trial, one machine, small scale; generality unproven; the 87%-vs-0% separation is far outside the ~20% noise floor, but the exact rate is one trial; the signed all-green analysis swapped the two models' rows; this claim uses the report and JSON attribution

Source: studies/05-cache-consistency/reports/20260921T-survey3.md · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

v2-16-strict-freshness-read-cost

external read copies / caches · single_run_directional

The strict-freshness contract cost nothing measurable in READ throughput: every controlled strict/relaxed pair differs by between -14.2% and +12.3%, inside the run's ~20% single-trial noise floor. The visible cost is on the write path (more fences), so this must never be restated as 'strict freshness is free'.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-06-claim12-full-noise-range](#), [conf-04-study05-db-only-cache](#)

Limits: read throughput only; one trial per measurement; the range is noise, not an error bar; small scale, closed loop

Source: studies/05-cache-consistency/reports/20260921T-survey3.md · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

5 Minor variants

CHAPTER · MINOR VARIANTS

A minor variant is one controlled change inside a family: index selection, full versus bounded embedding, trigger versus application maintenance, optimistic versus pessimistic concurrency, locks/CAS/isolation, pre-created versus on-demand inventory, cache-aside versus write-through, and strict versus relaxed freshness. Only the first and the concurrency variants have measured controlled pairs in the corpus; the rest are labelled with their gap or analogy status.

v2-01-normalized-index-first

normalized facts · single_run_directional

In the charity -> person -> donation tree, the first large read win is the query index: D2 indexed scores 13.3x D1 minimal's read score; flattening the foreign key instead (D3 flat+FK) reaches 20.5x.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-01-study01-d2d3-copied-key-package](#)

Limits: PostgreSQL 1 node and YugabyteDB 1/3 nodes in one run; small scale; one trial per cell; read score is a blended measure, not one query

Source: [studies/01-charity-tree/reports/20260912-small.md](#) · run 20260912-small · tag none (legacy)

v2-12-lock-vs-version-upper-bound

normalized facts · single_run_directional

Under one hot key with 16 writers a lock-before-update (c2_pessimistic_lock) delivered 1.76x the throughput of a version check (c1_optimistic_version), an UPPER BOUND on the lock's advantage because the run could not separate it from extra per-retry harness work; an unchecked read-modify-write (x1_lost_update_control) lost 93.7% of acknowledged updates.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one key, one installed product, 16 writers; PostgreSQL single node, small scale; one trial; -retries 1 control not run; 1.76x is one point on a writer curve, not a curve

Source: [studies/04-configuration-portal/reports/20260921T1215Z-small.md](#) · run 20260921T1215Z-small · tag run/04-configuration-portal/20260921T1215Z-small

v2-13-trigger-rollup-cost

rollups / materialized aggregates · single_run_directional

Maintaining the portal overview with a per-row trigger (n3_rollup_trigger) costs 6.85x on whole-configuration replacement (83.7 vs 573 ops/s, p99 2.5 -> 42 ms) while the application rollup (n4_rollup_app) keeps writes near the reference (593 ops/s) for the same ~15x read gain.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: PostgreSQL single node, small scale, one trial; the big effects (6.85x write, ~15x read) are above the run's ~1.7x read floor; smaller differences are not attributable; the trigger's product is a structural guarantee, not speed

Source: [studies/04-configuration-portal/reports/20260921T1215Z-small.md](#) · run 20260921T1215Z-small · tag run/04-configuration-portal/20260921T1215Z-small

v2-16-strict-freshness-read-cost

external read copies / caches · single_run_directional

The strict-freshness contract cost nothing measurable in READ throughput: every controlled strict/relaxed pair differs by between -14.2% and +12.3%, inside the run's ~20% single-trial noise floor. The visible cost is on the write path (more fences), so this must never be restated as 'strict freshness is free'.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-06-claim12-full-noise-range](#), [conf-04-study05-db-only-cache](#)

Limits: read throughput only; one trial per measurement; the range is noise, not an error bar; small scale, closed loop

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

6 Cross-family comparisons

CHAPTER · CROSS-FAMILY COMPARISONS

Comparisons here are controlled pairs or explicitly labelled analogy. No cross-study numeric ranking appears: no comparability record exists. The six confounds that bound every comparison are published beside the numbers, not in a footnote.

Coverage gap

Several family contrasts are family/package comparisons rather than one-decision pairs; see the confound register book/evidence/v2/confounds.json.

v2-03-embedding-not-ahead

embedded documents · single_run_directional

Embedding the donation child in the donor row (D6 embedded) never came out ahead of the normalized designs on reads: 2.4x D1 minimal's read score, below both indexed designs, with a 0.57x insert rate.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial; embedding here means a JSON/array child column, not a native document database

Source: studies/01-charity-tree/reports/20260912-small.md · run 20260912-small · tag none (legacy)

v2-11-section-seat-map

embedded documents · single_run_directional

On YugabyteDB the section document (L2 document) answers the 100k-seat map 4.5-11x faster than per-seat rows, while the section-sharded variant (L3 sharded) collapses point lookups to 26 ops/s; PostgreSQL shows no attributable difference. This is a family/package comparison, not a one-decision pair.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: conf-03-study03-layout-family-package

Limits: one read-only run, one trial, 5 s per measurement; quota-bound nodes; read-only phases; no control could fire; differences below ~1.8x are not attributable

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

7 Topologies

CHAPTER · TOPOLOGIES

Topology is a separate axis from design family: node count, replication, placement, endpoint routing, resource framing, network conditions and failure behaviour. The measured corpus covers PostgreSQL single-node and YugabyteDB single-node/3-node RF=3; placement and real-network coverage are gaps.

v2-gap-05-colocation-unverified

rolldown / duplicated data · gap

Physical data placement is a common open gap: no study verifies actual node placement. Study 01 D7/D24 and Study 03 L3 are placement intent with measured effects, not verified colocation; Study 02 records its own placement gap; Study 04 y1/y2 and Study 05's placement pair were never executed.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: containers on one laptop are not evidence of colocation; physical placement is undocumented to today's standard, not proven uncolocated

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

v2-gap-07-real-network-balanced-endpoints

normalized facts · gap

Every multi-node result runs on one laptop with no real network, and balanced client access across cluster endpoints is not proven. Multi-node ratios are placement experiments, not distributed-systems results.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: no real network between nodes; endpoint distribution across the cluster is not measured

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

v2-11-section-seat-map

embedded documents · single_run_directional

On YugabyteDB the section document (L2 document) answers the 100k-seat map 4.5-11x faster than per-seat rows, while the section-sharded variant (L3 sharded) collapses point lookups to 26 ops/s; PostgreSQL shows no attributable difference. This is a family/package comparison, not a one-decision pair.

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Confounds: [conf-03-study03-layout-family-package](#)

Limits: one read-only run, one trial, 5 s per measurement; quota-bound nodes; read-only phases; no control could fire; differences below ~1.8x are not attributable

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

8 Scenarios

CHAPTER · SCENARIOS

Scenarios group the studies by the reader’s domain: a charity donation tree, ticket overbooking and refunds, reserved seating and holds, a configuration portal, and an external cache. Each carries its own answerability questions and its own coverage gaps.

v2-06-hot-drop-designs

normalized facts · single_run_directional

For a hot drop, pre-created seats taken by compare-and-set or SKIP LOCKED (P3 CAS, P2 skip-locked) sell 4-8x faster than any single counter row, but only with retry backoff in mind, and a counter on the event row also blocks unrelated edits to it. Both deliberately wrong controls fired in the same run.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial per cell; the run has 42 cells of which 2 failed (C2 count (SER) on the two YugabyteDB topologies); those cells are not evidence; hot-drop workload only

Source: studies/02-ticket-booking/reports/20260913T021206Z.md · run 20260913T021206Z · tag run/02-ticket-booking/20260913T021206Z

v2-10-guarded-confirm-refusals

normalized facts · single_run_directional

On YugabyteDB a guarded confirmation is not enough: correct designs refused valid holds, and retrying the refused statement once removed every refusal at no measurable cost; on PostgreSQL the guarantee held everywhere.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: v1 matrix run, one trial per cell; the claim is about refusals, not throughput; retry-once is a design variant (S1r), not a tuning flag

Source: studies/03-reserved-seating/reports/20260915T002411Z.md · run 20260915T002411Z · tag run/03-reserved-seating/20260915T002411Z

v2-15-three-instance-staleness

external read copies / caches · mechanism_supported

Three logical application instances sharing one Redis cache on the write-through + relaxed strategy record about 87% wrong reads, while every single-instance relaxed cell records zero; a private in-memory lease cannot coordinate instances, and a database version token does not remove it. The 87.81% figure belongs to the OWNED model (o:opt:red:wth:rel); the legacy model’s cell is 87.24%.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: **conf-04-study05-db-only-cache**

Limits: single trial, one machine, small scale; generality unproven; the 87%-vs-0% separation is far outside the ~20% noise floor, but the exact rate is one trial; the signed all-green analysis swapped the two models’ rows; this claim uses the report and JSON attribution

Source: studies/05-cache-consistency/reports/20260921T-survey3.md · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

9 Evidence and reproduction

CHAPTER · EVIDENCE AND REPRODUCTION

Every number in this book resolves to a claim in `book/evidence/v2/claims.json`, and every claim to a measured cell in a cited report. This chapter explains the confidence cards, the run-tag provenance model, and how to reproduce a run in Podman.

9.1 The provenance model

Direct evidence

A run tag marks the producing code state, not a tree that contains the run's results. Check out the tag for the code, then read the committed results directory for the data.

v2-gap-06-native-datastore-families

embedded documents · gap

No native document, key-value, graph or wide-column database was measured: JSONB/state is not evidence for a native document database, and Redis was measured only as a cache. Cross-family transfers to native datastores are analogy, not evidence.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: do not transfer the embedded-document or cache results to a native family

Source: `studies/05-cache-consistency/reports/20260921T-survey3.md` · run `20260921T-survey3` · tag `run/05-cache-consistency/20260921T-survey3`

v2-gap-01-ledger-load-multiplier-unstable

append-only history / snapshots · gap

The append-only ledger's load-time multiplier is not published: the same pair and loader measured X1's pg-single load at 7.8x P3's in one run and 1.2x in the two sibling runs, with an empty harness diff.

Trials: 3 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: a single cell's one-shot load is not a design cost; requires its own repeated load-only cell before any multiplier is quoted

Source: `studies/02-ticket-booking/reports/20260920T234953Z.md` · run `20260920T234953Z` · tag `run/02-ticket-booking/20260920T234953Z`

9.2 Reproducing the book itself

The book is built in Podman from a pinned toolchain; the manifest records every input and the resulting hash.

```
book/build.sh          # builds the pinned image, compiles, verifies, writes the
                        manifest
cat book/dist/build-manifest.json
```

9.3 Reproducing a measured run

```
git checkout run/<study>/<run-id>    # the producing code state
infra/<topology>.sh up                # take the benchmark lock first
cd studies/<study> && ./run-<engine>.sh
```

A run's results live in `studies/<study>/results/<run-id>/`; the report beside them is generated, never hand-edited, and its inputs digest identifies the data.

10 Concepts

11 Information placement

CONCEPT · INFORMATION PLACEMENT

Where a fact lives decides what it can answer and what it costs to change. This concept frames the derive → index → materialize → copy/cache ladder and the answerability-first rule.

v2-07-refund-answerability-and-storage

append-only history / snapshots · single_run_directional

A ticketing database chosen only to prevent overbooking cannot report refunds at all: in 14 of 15 designs the refunds window is unanswerable and the sales/revenue and last-purchase reports are wrong across a refund. The append-only ledger (X1 CAS+ledger) that fixes it costs about 33-35% more storage.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: answerability is a schema fact, not a timing; the +33-35% storage delta is deterministic and stable across three runs; the P3 -> X1 read rows of this run are not attributable (cell-level effect) and are excluded

Source: studies/02-ticket-booking/reports/20260920T234953Z.md · run 20260920T234953Z · tag run/02-ticket-booking/20260920T234953Z

v2-02-rollup-family

rollups / materialized aggregates · single_run_directional

Parent rollups win the aggregate read score (D4 rollup/trg 73.8x and D5 rollup/app 69.9x against D1 minimal) and pay for it on inserts (2.6k/s and 2.1k/s against 6.2k/s).

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial per cell; PostgreSQL 1 node numbers shown; YugabyteDB in the same report; the write cost is per-insert on a fresh load

Source: studies/01-charity-tree/reports/20260912-small.md · run 20260912-small · tag none (legacy)

12 Indexes and hot rows

CONCEPT · INDEXES AND HOT ROWS

An index is the cheapest derived structure, but it is not free: it charges every write and it does not answer a question the schema cannot express. The hot-row case shows the arbitration bound.

v2-01-normalized-index-first

normalized facts · single_run_directional

In the charity -> person -> donation tree, the first large read win is the query index: D2 indexed scores 13.3x D1 minimal's read score; flattening the foreign key instead (D3 flat+FK) reaches 20.5x.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-01-study01-d2d3-copied-key-package](#)

Limits: PostgreSQL 1 node and YugabyteDB 1/3 nodes in one run; small scale; one trial per cell; read score is a blended measure, not one query

Source: [studies/01-charity-tree/reports/20260912-small.md](#) · run 20260912-small · tag none (legacy)

v2-06-hot-drop-designs

normalized facts · single_run_directional

For a hot drop, pre-created seats taken by compare-and-set or SKIP LOCKED (P3 CAS, P2 skip-locked) sell 4-8x faster than any single counter row, but only with retry backoff in mind, and a counter on the event row also blocks unrelated edits to it. Both deliberately wrong controls fired in the same run.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial per cell; the run has 42 cells of which 2 failed (C2 count (SER) on the two YugabyteDB topologies); those cells are not evidence; hot-drop workload only

Source: [studies/02-ticket-booking/reports/20260913T021206Z.md](#) · run 20260913T021206Z · tag run/02-ticket-booking/20260913T021206Z

13 Concurrency control

CONCEPT · CONCURRENCY CONTROL

Optimistic (version-checked) and pessimistic (lock-before-update) strategies under the same business invariant. Both local engine mechanisms and application retry loops belong here.

v2-12-lock-vs-version-upper-bound

normalized facts · single_run_directional

Under one hot key with 16 writers a lock-before-update (c2_pessimistic_lock) delivered 1.76x the throughput of a version check (c1_optimistic_version), an UPPER BOUND on the lock's advantage because the run could not separate it from extra per-retry harness work; an unchecked read-modify-write (x1_lost_update_control) lost 93.7% of acknowledged updates.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one key, one installed product, 16 writers; PostgreSQL single node, small scale; one trial; -retries 1 control not run; 1.76x is one point on a writer curve, not a curve

Source: studies/04-configuration-portal/reports/20260921T1215Z-small.md · run 20260921T1215Z-small · tag run/04-configuration-portal/20260921T1215Z-small

14 Derived state and history

CONCEPT · DERIVED STATE AND HISTORY

Rollups, duplicated keys and append-only histories are all derived state with different maintenance costs and different answerability. History is the only one that can answer a question after the event was undone.

v2-13-trigger-rollup-cost

rollups / materialized aggregates · single_run_directional

Maintaining the portal overview with a per-row trigger (n3_rollup_trigger) costs 6.85x on whole-configuration replacement (83.7 vs 573 ops/s, p99 2.5 -> 42 ms) while the application rollup (n4_rollup_app) keeps writes near the reference (593 ops/s) for the same ~15x read gain.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: PostgreSQL single node, small scale, one trial; the big effects (6.85x write, ~15x read) are above the run's ~1.7x read floor; smaller differences are not attributable; the trigger's product is a structural guarantee, not speed

Source: studies/04-configuration-portal/reports/20260921T1215Z-small.md · run 20260921T1215Z-small · tag run/04-configuration-portal/20260921T1215Z-small

v2-09-postgres-ledger-ratio-unsettled

append-only history / snapshots · single_run_directional

The two arms of the P3/X1 pair disagree in sign on PostgreSQL (32 buyers with P3 first: X1 faster; 128 buyers with X1 first: P3 faster), and the 32-buyer arm's own P3 trials disagree by 92-283%. No PostgreSQL ledger-cost ratio is quotable from this study; the run's own YugabyteDB range is 1.7-5.7x (not the headline's narrower 1.7-2.1x).

Trials: 1 whole-run replications, 3 fresh-load trials per cell, 32 inner iterations.

Confounds: [conf-05-study02-buyer-arms-disagree](#)

Limits: buyer count and design order changed together, so neither can be isolated; one arm is a partial cell on yb-cluster3 (two-trial medians); a repeated-design interleaved control is required before any PostgreSQL ratio is quoted

Source: studies/02-ticket-booking/reports/20260921T183722Z.md · run 20260921T183722Z · tag run/02-ticket-booking/20260921T183722Z

15 Expiry and clock authority

CONCEPT · EXPIRY AND CLOCK AUTHORITY

Expiry is a write that no client issues; whose clock decides matters. The corpus exposes the difference but cannot answer the funnel that would size a hold window.

v2-10-guarded-confirm-refusals

normalized facts · single_run_directional

On YugabyteDB a guarded confirmation is not enough: correct designs refused valid holds, and retrying the refused statement once removed every refusal at no measurable cost; on PostgreSQL the guarantee held everywhere.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: v1 matrix run, one trial per cell; the claim is about refusals, not throughput; retry-once is a design variant (S1r), not a tuning flag

Source: studies/03-reserved-seating/reports/20260915T002411Z.md · run 20260915T002411Z · tag run/03-reserved-seating/20260915T002411Z

v2-gap-02-hold-funnel-unanswerable

append-only history / snapshots · gap

No Study 03 design can answer the hold funnel report that would size the 40-minute hold window: releasing an expired hold clears the row, so an expired hold is indistinguishable from a seat never held.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: a schema limitation, not a missing measurement

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

16 Cache consistency

CONCEPT · CACHE CONSISTENCY

A cache is a read copy with a freshness contract. Invalidation is a fence, not a deletion; publication must be refused when the fence moved. A private in-memory lease does not coordinate instances sharing a store.

v2-14-cache-throughput-gain

external read copies / caches · single_run_directional

Adding a cache to warm cacheable reads buys 2.2-3.5x in throughput and an order of magnitude in p99 (15.8-27.1 ms to 0.8-2.0 ms). This is an add-cache result under resource_framing db-only, not equal-total resource efficiency, and Redis here is a shared cache, never an authoritative key-value store.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-04-study05-db-only-cache](#)

Limits: one trial per measurement, one machine, closed loop; small scale (800 donors), PostgreSQL + Redis cache only; warm cacheable reads only; the Redis container's budget is outside the comparison (gap); no YugabyteDB, cluster, verified placement or open-loop demand

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · [run 20260921T-survey3](#) · [tag run/05-cache-consistency/20260921T-survey3](#)

v2-15-three-instance-staleness

external read copies / caches · mechanism_supported

Three logical application instances sharing one Redis cache on the write-through + relaxed strategy record about 87% wrong reads, while every single-instance relaxed cell records zero; a private in-memory lease cannot coordinate instances, and a database version token does not remove it. The 87.81% figure belongs to the OWNED model (o:opt:red:wth:rel); the legacy model's cell is 87.24%.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-04-study05-db-only-cache](#)

Limits: single trial, one machine, small scale; generality unproven; the 87%-vs-0% separation is far outside the ~20% noise floor, but the exact rate is one trial; the signed all-green analysis swapped the two models' rows; this claim uses the report and JSON attribution

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · [run 20260921T-survey3](#) · [tag run/05-cache-consistency/20260921T-survey3](#)

v2-16-strict-freshness-read-cost

external read copies / caches · single_run_directional

The strict-freshness contract cost nothing measurable in READ throughput: every controlled strict/relaxed pair differs by between -14.2% and +12.3%, inside the run's ~20% single-trial noise floor. The visible cost is on the write path (more fences), so this must never be restated as 'strict freshness is free'.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-06-claim12-full-noise-range](#), [conf-04-study05-db-only-cache](#)

Limits: read throughput only; one trial per measurement; the range is noise, not an error bar; small scale, closed loop

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · [run 20260921T-survey3](#) · [tag run/05-cache-consistency/20260921T-survey3](#)

17 Distributed placement

CONCEPT · DISTRIBUTED PLACEMENT

Node count, replication and actual data placement are separate facts. The corpus measures node count and replication but does not verify placement or a real network.

v2-gap-05-colocation-unverified

rolldown / duplicated data · gap

Physical data placement is a common open gap: no study verifies actual node placement. Study 01 D7/D24 and Study 03 L3 are placement intent with measured effects, not verified colocation; Study 02 records its own placement gap; Study 04 y1/y2 and Study 05's placement pair were never executed.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: containers on one laptop are not evidence of colocation; physical placement is undocumented to today's standard, not proven uncolocated

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

v2-gap-07-real-network-balanced-endpoints

normalized facts · gap

Every multi-node result runs on one laptop with no real network, and balanced client access across cluster endpoints is not proven. Multi-node ratios are placement experiments, not distributed-systems results.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: no real network between nodes; endpoint distribution across the cluster is not measured

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

18 Reading benchmarks honestly

CONCEPT · READING BENCHMARKS

Closed vs open loop, single vs repeated trials, per-node vs equal-total framing, throttled cores, and the difference between a design effect and a cell-order effect. This concept is the reader’s guard against quoting a number the run cannot support.

v2-16-strict-freshness-read-cost

external read copies / caches · single_run_directional

The strict-freshness contract cost nothing measurable in READ throughput: every controlled strict/relaxed pair differs by between -14.2% and +12.3%, inside the run's ~20% single-trial noise floor. The visible cost is on the write path (more fences), so this must never be restated as 'strict freshness is free'.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-06-claim12-full-noise-range](#), [conf-04-study05-db-only-cache](#)

Limits: read throughput only; one trial per measurement; the range is noise, not an error bar; small scale, closed loop

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

v2-gap-01-ledger-load-multiplier-unstable

append-only history / snapshots · gap

The append-only ledger's load-time multiplier is not published: the same pair and loader measured X1's pg-single load at 7.8x P3's in one run and 1.2x in the two sibling runs, with an empty harness diff.

Trials: 3 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: a single cell's one-shot load is not a design cost; requires its own repeated load-only cell before any multiplier is quoted

Source: [studies/02-ticket-booking/reports/20260920T234953Z.md](#) · run 20260920T234953Z · tag run/02-ticket-booking/20260920T234953Z

v2-gap-04-study05-unmeasured-regimes

external read copies / caches · gap

Real-TTL churn, medium scale, repeated trials, cache resource accounting, YugabyteDB/cluster, verified placement and open-loop demand are unmeasured for Study 05.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Confounds: [conf-04-study05-db-only-cache](#)

Limits: owned by task-20260922T025912Z-study05-v2-protocol

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

19 Evidence registry (active v2 claims)

The active registry holds 23 claims. This index is rendered from the registry file at build time; the digest above identifies its content.

19.1 normalized facts

v2-01-normalized-index-first

normalized facts · single_run_directional

In the charity -> person -> donation tree, the first large read win is the query index: D2 indexed scores 13.3x D1 minimal's read score; flattening the foreign key instead (D3 flat+FK) reaches 20.5x.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-01-study01-d2d3-copied-key-package](#)

Limits: PostgreSQL 1 node and YugabyteDB 1/3 nodes in one run; small scale; one trial per cell; read score is a blended measure, not one query

Source: [studies/01-charity-tree/reports/20260912-small.md](#) · run 20260912-small · tag none (legacy)

v2-06-hot-drop-designs

normalized facts · single_run_directional

For a hot drop, pre-created seats taken by compare-and-set or SKIP LOCKED (P3 CAS, P2 skip-locked) sell 4-8x faster than any single counter row, but only with retry backoff in mind, and a counter on the event row also blocks unrelated edits to it. Both deliberately wrong controls fired in the same run.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial per cell; the run has 42 cells of which 2 failed (C2 count (SER) on the two YugabyteDB topologies); those cells are not evidence; hot-drop workload only

Source: [studies/02-ticket-booking/reports/20260913T021206Z.md](#) · run 20260913T021206Z · tag run/02-ticket-booking/20260913T021206Z

v2-10-guarded-confirm-refusals

normalized facts · single_run_directional

On YugabyteDB a guarded confirmation is not enough: correct designs refused valid holds, and retrying the refused statement once removed every refusal at no measurable cost; on PostgreSQL the guarantee held everywhere.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: v1 matrix run, one trial per cell; the claim is about refusals, not throughput; retry-once is a design variant (S1r), not a tuning flag

Source: [studies/03-reserved-seating/reports/20260915T002411Z.md](#) · run 20260915T002411Z · tag run/03-reserved-seating/20260915T002411Z

v2-12-lock-vs-version-upper-bound

normalized facts · single_run_directional

Under one hot key with 16 writers a lock-before-update (c2_pessimistic_lock) delivered 1.76x the throughput of a version check (c1_optimistic_version), an UPPER BOUND on the lock's advantage because the run could not separate it from extra per-retry harness work; an unchecked read-modify-write (x1_lost_update_control) lost 93.7% of acknowledged updates.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one key, one installed product, 16 writers; PostgreSQL single node, small scale; one trial; -retries 1 control not run; 1.76x is one point on a writer curve, not a curve

Source: [studies/04-configuration-portal/reports/20260921T1215Z-small.md](#) · run 20260921T1215Z-small · tag run/04-configuration-portal/20260921T1215Z-small

v2-gap-03-study04-missing-designs

normalized facts · gap

Eight of the eighteen planned Study 04 configuration-portal designs are not implemented, and cardinality, cadence/churn, repeated randomized trials, equal-total resources, placement and YugabyteDB are unmeasured. These are coverage gaps, not null results.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: no conclusion may read 'the missing designs would not change this'; owned by task-20260922T025912Z-study04-v2-protocol

Source: studies/04-configuration-portal/reports/20260921T1215Z-small.md · run 20260921T1215Z-small · tag run/04-configuration-portal/20260921T1215Z-small

v2-gap-07-real-network-balanced-endpoints

normalized facts · gap

Every multi-node result runs on one laptop with no real network, and balanced client access across cluster endpoints is not proven. Multi-node ratios are placement experiments, not distributed-systems results.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: no real network between nodes; endpoint distribution across the cluster is not measured

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

19.2 rolldown / duplicated data

v2-05-d3-read-advantage-repeated

rolldown / duplicated data · repeated_controlled

D3's (d3_flattened_fk) read advantage over the normalized indexed design survives five fresh loads; the recency SQL and actual heap avoidance explain the gain, while its insert cost remains imprecise.

Trials: 1 whole-run replications, 5 fresh-load trials per cell, 1 inner iterations.

Limits: five fresh-load trials per cell, one session; the insert-cost estimate is imprecise and must not be quoted as a ratio

Source: studies/01-charity-tree/reports/20260913T125342Z-v3.md · run 20260913T125342Z-v3 · tag run/01-charity-tree/20260913T125342Z-v3

v2-gap-05-colocation-unverified

rolldown / duplicated data · gap

Physical data placement is a common open gap: no study verifies actual node placement. Study 01 D7/D24 and Study 03 L3 are placement intent with measured effects, not verified colocation; Study 02 records its own placement gap; Study 04 y1/y2 and Study 05's placement pair were never executed.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: containers on one laptop are not evidence of colocation; physical placement is undocumented to today's standard, not proven uncolocated

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

19.3 rollups / materialized aggregates

v2-02-rollup-family

rollups / materialized aggregates · single_run_directional

Parent rollups win the aggregate read score (D4 rollup/trg 73.8x and D5 rollup/app 69.9x against D1 minimal) and pay for it on inserts (2.6k/s and 2.1k/s against 6.2k/s).

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial per cell; PostgreSQL 1 node numbers shown; YugabyteDB in the same report; the write cost is per-insert on a fresh load

Source: studies/01-charity-tree/reports/20260912-small.md · run 20260912-small · tag none (legacy)

v2-04-recency-maintained-index

rollups / materialized aggregates · single_run_directional

The recency flag works, but an index on the rollup column answers the same question faster: comparing `d20_recency_flag` with `d22_recency_rollup_idx` in the same run gives the reported read speedup on PostgreSQL and YugabyteDB.

Trials: 1 whole-run replications, 3 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-02-study01-d20d22-write-axis](#)

Limits: the write comparison is confounded: d22 also maintains D4's multi-column rollup package, so its write cost is not the index alone; three trials per cell; claim-02's v1 trials field was recorded as 1 and is corrected here

Source: [studies/01-charity-tree/reports/20260916T090036Z-v3.md](#) · [run 20260916T090036Z-v3](#) · [tag run/01-charity-tree/20260916T090036Z-v3](#)

v2-13-trigger-rollup-cost

rollups / materialized aggregates · single_run_directional

Maintaining the portal overview with a per-row trigger (`n3_rollup_trigger`) costs 6.85x on whole-configuration replacement (83.7 vs 573 ops/s, p99 2.5 -> 42 ms) while the application rollup (`n4_rollup_app`) keeps writes near the reference (593 ops/s) for the same ~15x read gain.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: PostgreSQL single node, small scale, one trial; the big effects (6.85x write, ~15x read) are above the run's ~1.7x read floor; smaller differences are not attributable; the trigger's product is a structural guarantee, not speed

Source: [studies/04-configuration-portal/reports/20260921T1215Z-small.md](#) · [run 20260921T1215Z-small](#) · [tag run/04-configuration-portal/20260921T1215Z-small](#)

19.4 embedded documents

v2-03-embedding-not-ahead

embedded documents · single_run_directional

Embedding the donation child in the donor row (D6 embedded) never came out ahead of the normalized designs on reads: 2.4x D1 minimal's read score, below both indexed designs, with a 0.57x insert rate.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: one run, one trial; embedding here means a JSON/array child column, not a native document database

Source: [studies/01-charity-tree/reports/20260912-small.md](#) · [run 20260912-small](#) · [tag none](#) (legacy)

v2-11-section-seat-map

embedded documents · single_run_directional

On YugabyteDB the section document (L2 document) answers the 100k-seat map 4.5-11x faster than per-seat rows, while the section-sharded variant (L3 sharded) collapses point lookups to 26 ops/s; PostgreSQL shows no attributable difference. This is a family/package comparison, not a one-decision pair.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-03-study03-layout-family-package](#)

Limits: one read-only run, one trial, 5 s per measurement; quota-bound nodes; read-only phases; no control could fire; differences below ~1.8x are not attributable

Source: [studies/03-reserved-seating/reports/20260921T231328Z.md](#) · [run 20260921T231328Z](#) · [tag run/03-reserved-seating/20260921T231328Z](#)

v2-gap-06-native-datastore-families

embedded documents · gap

No native document, key-value, graph or wide-column database was measured: JSONB/state is not evidence for a native document database, and Redis was measured only as a cache. Cross-family transfers to native datastores are analogy, not evidence.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: do not transfer the embedded-document or cache results to a native family

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · [run 20260921T-survey3](#) · [tag run/05-cache-consistency/20260921T-survey3](#)

19.5 append-only history / snapshots

v2-07-refund-answerability-and-storage

append-only history / snapshots · single_run_directional

A ticketing database chosen only to prevent overbooking cannot report refunds at all: in 14 of 15 designs the refunds window is unanswerable and the sales/revenue and last-purchase reports are wrong across a refund. The append-only ledger (X1 CAS+ledger) that fixes it costs about 33-35% more storage.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: answerability is a schema fact, not a timing; the +33-35% storage delta is deterministic and stable across three runs; the P3 -> X1 read rows of this run are not attributable (cell-level effect) and are excluded

Source: studies/02-ticket-booking/reports/20260920T234953Z.md · run 20260920T234953Z · tag run/02-ticket-booking/20260920T234953Z

v2-08-ledger-race-cost-128-buyers

append-only history / snapshots · single_run_directional

At 128 buyers the ledger costs 1.1-2.3x of the sell-out race on PostgreSQL and 1.8-2.2x on YugabyteDB with tight spreads, 2.1-2.5x on the refund itself, and not one under-booked seat anywhere; it delays the organiser's own edit by up to 5x.

Trials: 1 whole-run replications, 3 fresh-load trials per cell, 128 inner iterations.

Confounds: [conf-05-study02-buyer-arms-disagree](#)

Limits: one small run, closed loop, quota-bound nodes; scoped to 128 buyers; must not be extrapolated to 32 buyers; single run, three trials per tier

Source: studies/02-ticket-booking/reports/20260921T205212Z.md · run 20260921T205212Z · tag run/02-ticket-booking/20260921T205212Z

v2-09-postgres-ledger-ratio-unsettled

append-only history / snapshots · single_run_directional

The two arms of the P3/X1 pair disagree in sign on PostgreSQL (32 buyers with P3 first: X1 faster; 128 buyers with X1 first: P3 faster), and the 32-buyer arm's own P3 trials disagree by 92-283%. No PostgreSQL ledger-cost ratio is quotable from this study; the run's own YugabyteDB range is 1.7-5.7x (not the headline's narrower 1.7-2.1x).

Trials: 1 whole-run replications, 3 fresh-load trials per cell, 32 inner iterations.

Confounds: [conf-05-study02-buyer-arms-disagree](#)

Limits: buyer count and design order changed together, so neither can be isolated; one arm is a partial cell on yb-cluster3 (two-trial medians); a repeated-design interleaved control is required before any PostgreSQL ratio is quoted

Source: studies/02-ticket-booking/reports/20260921T183722Z.md · run 20260921T183722Z · tag run/02-ticket-booking/20260921T183722Z

v2-gap-01-ledger-load-multiplier-unstable

append-only history / snapshots · gap

The append-only ledger's load-time multiplier is not published: the same pair and loader measured X1's pg-single load at 7.8x P3's in one run and 1.2x in the two sibling runs, with an empty harness diff.

Trials: 3 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Limits: a single cell's one-shot load is not a design cost; requires its own repeated load-only cell before any multiplier is quoted

Source: studies/02-ticket-booking/reports/20260920T234953Z.md · run 20260920T234953Z · tag run/02-ticket-booking/20260920T234953Z

v2-gap-02-hold-funnel-unanswerable

append-only history / snapshots · gap

No Study 03 design can answer the hold funnel report that would size the 40-minute hold window: releasing an expired hold clears the row, so an expired hold is indistinguishable from a seat never held.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Limits: a schema limitation, not a missing measurement

Source: studies/03-reserved-seating/reports/20260921T231328Z.md · run 20260921T231328Z · tag run/03-reserved-seating/20260921T231328Z

19.6 external read copies / caches

v2-14-cache-throughput-gain

external read copies / caches · single_run_directional

Adding a cache to warm cacheable reads buys 2.2-3.5x in throughput and an order of magnitude in p99 (15.8-27.1 ms to 0.8-2.0 ms). This is an add-cache result under resource_framing db-only, not equal-total resource efficiency, and Redis here is a shared cache, never an authoritative key-value store.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-04-study05-db-only-cache](#)

Limits: one trial per measurement, one machine, closed loop; small scale (800 donors), PostgreSQL + Redis cache only; warm cacheable reads only; the Redis container's budget is outside the comparison (gap); no YugabyteDB, cluster, verified placement or open-loop demand

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

v2-15-three-instance-staleness

external read copies / caches · mechanism_supported

Three logical application instances sharing one Redis cache on the write-through + relaxed strategy record about 87% wrong reads, while every single-instance relaxed cell records zero; a private in-memory lease cannot coordinate instances, and a database version token does not remove it. The 87.81% figure belongs to the OWNED model (o:opt:red:wth:rel); the legacy model's cell is 87.24%.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-04-study05-db-only-cache](#)

Limits: single trial, one machine, small scale; generality unproven; the 87%-vs-0% separation is far outside the ~20% noise floor, but the exact rate is one trial; the signed all-green analysis swapped the two models' rows; this claim uses the report and JSON attribution

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

v2-16-strict-freshness-read-cost

external read copies / caches · single_run_directional

The strict-freshness contract cost nothing measurable in READ throughput: every controlled strict/relaxed pair differs by between -14.2% and +12.3%, inside the run's ~20% single-trial noise floor. The visible cost is on the write path (more fences), so this must never be restated as 'strict freshness is free'.

Trials: 1 whole-run replications, 1 fresh-load trials per cell, 1 inner iterations.

Confounds: [conf-06-claim12-full-noise-range](#), [conf-04-study05-db-only-cache](#)

Limits: read throughput only; one trial per measurement; the range is noise, not an error bar; small scale, closed loop

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

v2-gap-04-study05-unmeasured-regimes

external read copies / caches · gap

Real-TTL churn, medium scale, repeated trials, cache resource accounting, YugabyteDB/cluster, verified placement and open-loop demand are unmeasured for Study 05.

Trials: 0 whole-run replications, 0 fresh-load trials per cell, 0 inner iterations.

Confounds: [conf-04-study05-db-only-cache](#)

Limits: owned by task-20260922T025912Z-study05-v2-protocol

Source: [studies/05-cache-consistency/reports/20260921T-survey3.md](#) · run 20260921T-survey3 · tag run/05-cache-consistency/20260921T-survey3

19.7 Retired v1 claims

- [claim-gap-04-study02-03-validation-open](#) — The validation it describes as open was completed and integrated 2026-09-22 (task-20260922T025912Z-study02-03-v2-validation). Not carried into v2.